

A First Categorical DWLS Profile-Hessian Check

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Purpose

This note records a first diagnostic exploration of the regular profile-Hessian roadmap in a genuinely categorical DWLS setting. The goal is not a finite-sample simulation yet. The goal is to check whether the categorical DWLS weight channel can visibly change the nested-test reference law once we leave exact fit.

The companion script is

`docs/research/notes/ordinal_dwls_profile_check.R.`

1 Setup

The population is a four-variable binary ordinal latent-response model. The first-stage object is

$$x = (u, \gamma),$$

where u stacks the four thresholds and six tetrachoric correlations, and

$$\gamma = \text{diag}(\Gamma_u)$$

is the diagonal of the asymptotic covariance of u . DWLS uses $W = \text{diag}(\gamma)^{-1}$. The script computes u , γ , and the joint covariance of (u, γ) by finite-difference delta method from the 2^4 binary cell probabilities.

The nested pair is:

A : congeneric one-factor latent-response correlations, B : tau-equivalent one-factor latent-response

Thresholds are free in both models. Thus the nested contrast tests equality of the loading structure, not threshold equality.

For the nested contrast B versus A , the script compares three quadratic reference laws:

- U_{GN} : Gauss–Newton fixed-weight projector, using only u ,
- Q_{fixed} : fixed-weight profile Hessian in u ,
- Q_{full} : full profile Hessian in $x = (u, \gamma)$.

The last object is the categorical DWLS version of the general $Q_B - Q_A$ roadmap object.

2 Results

Scenario	$\sqrt{2f_A}$	$\sqrt{2f_B}$	$\text{tr}(Q_{\text{fixed}}\Gamma_u)$	$\text{tr}(Q_{\text{full}}\Gamma_x)$
Exact nested null	0.0000	0.0000	2.8925	2.8924
Near-null misspec., $\epsilon = 0.04$	0.0499	0.0499	2.9109	2.9391
Near-null misspec., $\epsilon = 0.08$	0.1004	0.1004	2.9708	3.1582
Restriction false, congeneric	0.0000	0.1260	2.9116	3.0915

The corresponding full/fixed trace ratios were

$$1.0000, \quad 1.0097, \quad 1.0631, \quad 1.0618.$$

The exact nested null behaves as expected: the residual is essentially zero, so the DWLS weight channel is dormant and the full profile law collapses onto the fixed-weight law. Under fixed misspecification, the full DWLS law separates from the fixed-weight law. In the two near-null misspecification directions, the trace increases by about 1.0% and 6.3%.

The eigenvalue movement is more informative than the trace alone. For $\epsilon = 0.08$, the leading eigenvalues changed from

$$(1.2947, \ 0.9386, \ 0.7376)$$

under the fixed-weight profile Hessian to

$$(1.5551, \ 0.9324, \ 0.6709)$$

under the full DWLS profile Hessian. So the weight channel does not merely rescale the law; it reshapes it.

3 An Exact Symmetry Pseudo-Null

The unequal-threshold rows above are useful diagnostics, but they are only near-null examples. A cleaner categorical pseudo-null can be obtained by symmetry. Set all binary thresholds to zero and start from the tau-equivalent latent-response correlation matrix with loading $\lambda = 0.55$, so every off-diagonal correlation is $\lambda^2 = 0.3025$. Then add local dependence on the four edges of a cycle:

$$R_{12} = R_{14} = R_{23} = R_{34} = 0.3025 + \epsilon, \quad R_{13} = R_{24} = 0.3025.$$

All off-diagonal correlations remain nonzero, but the population has two pair orbits: adjacent pairs and opposite pairs. The four-cycle is vertex-transitive. With equal thresholds, this symmetry forces the congeneric pseudo-true loadings and thresholds to be equal, so the larger model's pseudo-true point lies in the tau-equivalent submodel even though both models are misspecified when $\epsilon > 0$.

Numerically, across the grid below the value difference satisfies $|f_B - f_A| < 6 \times 10^{-14}$ and the profile-score difference is below 2.1×10^{-7} . The ordinary nested reference has three degrees of freedom, with $q_{0.95}(\chi_3^2) = 7.8147$.

ϵ	$\sqrt{2f_A}$	$\text{tr}(Q_{\text{fixed}}\Gamma_u)$	$\text{tr}(Q_{\text{full}}\Gamma_x)$	$q_{0.95}^{\text{full}}$	$\Pr_{\text{full}}\{T > q_{0.95}(\chi_3^2)\}$
0.02	0.0158	2.8544	2.8577	7.449	0.042
0.04	0.0317	2.8475	2.8606	7.471	0.043
0.06	0.0477	2.8408	2.8510	7.476	0.043
0.08	0.0638	2.8331	2.8842	7.585	0.045
0.10	0.0801	2.8234	2.8959	7.655	0.047

The effect is smaller than in the unequal-threshold near-null direction, but it is cleaner: the pseudo-null is protected by symmetry, the residual is nonzero, and the γ block still changes the full profile law. At $\epsilon = 0.10$, the full DWLS eigenvalues are approximately

$$(1.1151, 1.1151, 0.6656),$$

whereas the fixed-weight trace is still below three. The ordinary χ_3^2 cutoff is mildly conservative here rather than catastrophic. That is a good sign for this as a realistic simulation design: the point is not to manufacture a giant failure, but to quantify how much the estimated categorical weight matters at plausible levels of limited-information misfit.

4 Interpretation

This is consistent with the general theory. The DWLS profile score has two blocks:

$$s_u = -Wr, \quad (s_\gamma)_k = -\frac{1}{2} \frac{r_k^2}{\gamma_k^2}.$$

The γ block is residual-driven. It vanishes at exact fit and becomes a leading component under fixed misspecification. This is exactly why categorical DWLS is a strong applied example for the paper: the most common ordinal SEM workflow uses an estimated diagonal weight, and that weight is asymptotically irrelevant only in the exact-fit regime.

5 Caveats

The first near-null misspecification scenarios were constructed to keep the congeneric and tau-equivalent fits nearly tied, but they are not exact pseudo-null designs. The four-cycle construction fixes that problem for a single-group binary CFA example. It is still only a population calculation: finite-sample rejection rates have not yet been simulated.

The script is intentionally small and finite-difference based. It is useful for orientation, not for final numerical claims. A production version should reuse the analytic ordinal moment influence and DWLS weight machinery in `magmaan`.

6 Next Steps

1. Run finite-sample simulations for the four-cycle pseudo-null, comparing ordinary χ^2 , fixed-weight Q_{fixed} , and full DWLS Q_{full} references.
2. Search for larger but still plausible symmetry-protected categorical designs by varying the number of indicators, thresholds, categories, and local dependence graph.
3. Repeat for multigroup threshold/loading invariance, where categorical DWLS is most common in practice.
4. Port the profile-Hessian calculation to `magmaan` using the existing ordinal moment and weight caches.